



Vitality Kitchen

Meal Planner Application

PROJECT DOCUMENTATION REPORT

DATE

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DEVELOPMENT TEAM

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Project Overview

Introduction

Validity Kitchen is a full-stack, MERN-based meal planning and recipe management web application designed to help users take control of their nutrition and weekly meal scheduling. Developed as an academic capstone project at Ain Shams University — Faculty of Computer and Information Sciences (2026), the platform eliminates the daily friction of deciding what to eat by offering a centralized hub for discovering healthy, macro-calculated recipes, organizing meals week by week, and automatically generating shopping lists.

Project Objectives

- 1. Provide a secure, JWT-authenticated environment for personal meal planning
- 2. Enable users to browse, create, edit, and rate recipes with automatic nutritional calculation
- 3. Allow users to plan Breakfast, Lunch, and Dinner for every day of any given week
- 4. Auto-generate an aggregated weekly grocery/shopping list from planned meals
- 5. Offer a personalized profile with dietary preference management and usage statistics
- 6. Expose a well-documented REST API with Swagger/OpenAPI interactive documentation

Project Scope

- User registration, authentication & profile management
- Recipe CRUD with Base64 image upload
- Automatic macro-nutrition calculation per recipe and per serving
- 5-star community recipe rating system
- Weekly meal planner (Mon–Sun, Breakfast/Lunch/Dinner)
- Grocery list generation from active meal plans
- Favorites management
- Dietary preference tagging (Vegan, Keto, Gluten-Free, etc.)
- RESTful backend API with JWT authentication
- Swagger/OpenAPI interactive docs at /api-docs
- 27 backend tests (Jest + Supertest)

Entities & Roles

■ Guest User	Can browse all public recipes (read-only). Cannot access meal planning, favorites, or profile features. Must register to gain full access.
■ Registered User	Full access to all application features. Owns personal recipes, meal plans, and favorites. Can set dietary preferences and a daily calorie goal. Can rate any recipe (one rating per user per recipe).
■■ REST API (Backend)	Acts as the single source of truth between frontend and database. Enforces authentication via JWT Bearer tokens on all write operations. Handles all business logic.
■■ MongoDB Database	Persists all application data: users, recipes, meal plans. Enforces compound unique index on (user, startDate) in MealPlan to prevent duplicates.